DO NOW — RECALL

# Lesson 097 · Little Man Computer: Branches and Loops

Name: _____ Class: _____ Date: _____

**1. What does BRA do in LMC?**

**2. When does BRZ branch?**

**3. What does LDA COUNT do?**

**4. Stack Push2,Push5,Pop,Push8: final bottom-to-top?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Branches unconditionally to a label.

### 2. Model answer

When accumulator equals zero.

### 3. Model answer

Loads COUNT's mailbox contents into ACC.

### 4. Model answer

[2,8].

### By the end, you can…

• BRA always changes which instruction is fetched next; BRZ branches only when ACC is zero.

• The loop's memory variable COUNT changes after SUB ONE and STA COUNT. Reload it with LDA COUNT at each iteration.

• Trace ACC, COUNT, branch decision and outputs. Stopping before OUT when COUNT=0 prevents printing zero.

### Vocabulary — know these terms

BRA — Always branch to label.

BRZ — Branch if ACC zero.

label — Symbolic instruction/memory address.

loop — Repeated control flow.

termination — Stop condition for loop.

NEW LEARNING

# Read, annotate and explain

## Observe

BRA always changes which instruction is fetched next; BRZ branches only when ACC is zero.

## Apply

The loop's memory variable COUNT changes after SUB ONE and STA COUNT. Reload it with LDA COUNT at each iteration.

## Evaluate

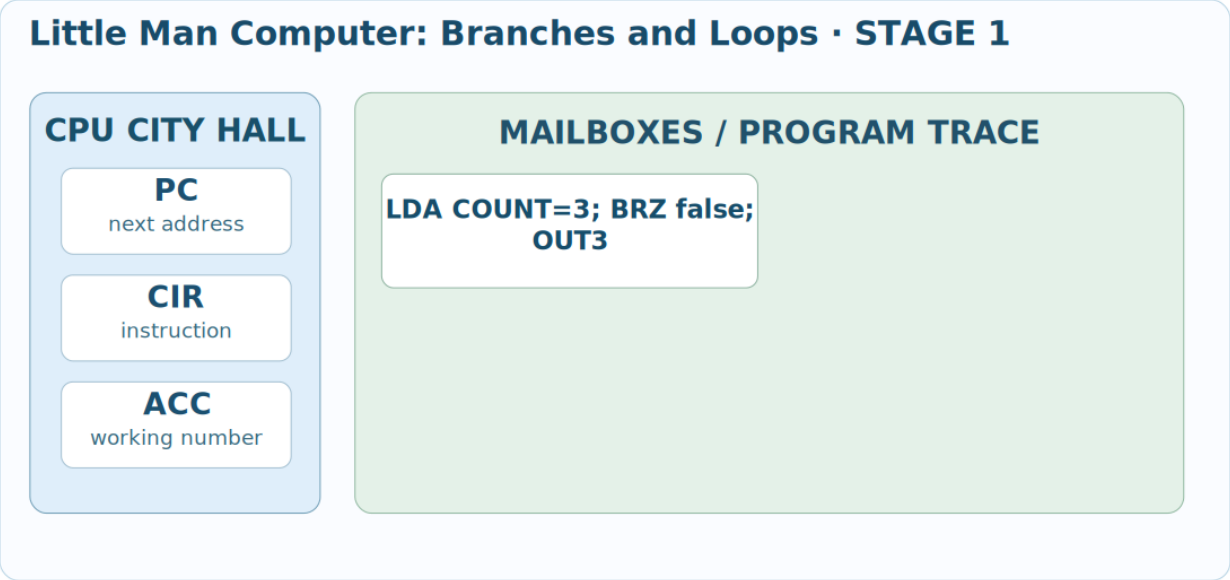
Trace ACC, COUNT, branch decision and outputs. Stopping before OUT when COUNT=0 prevents printing zero.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

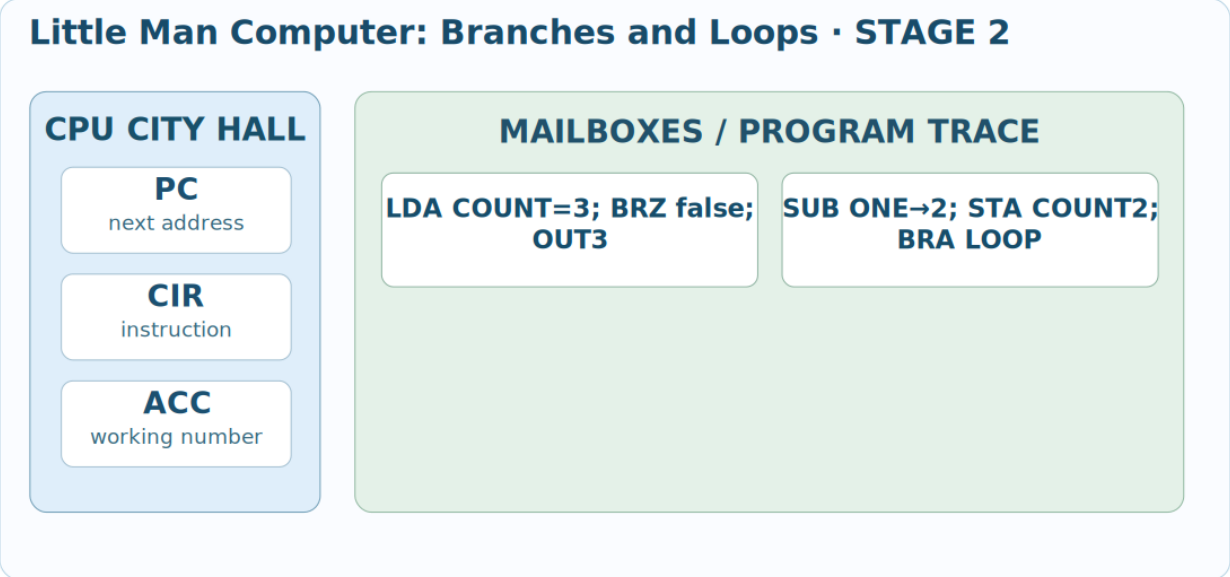
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Little Man Computer: Branches and Loops · STAGE 1; LDA COUNT=3; BRZ false; OUT3

**New Learning**

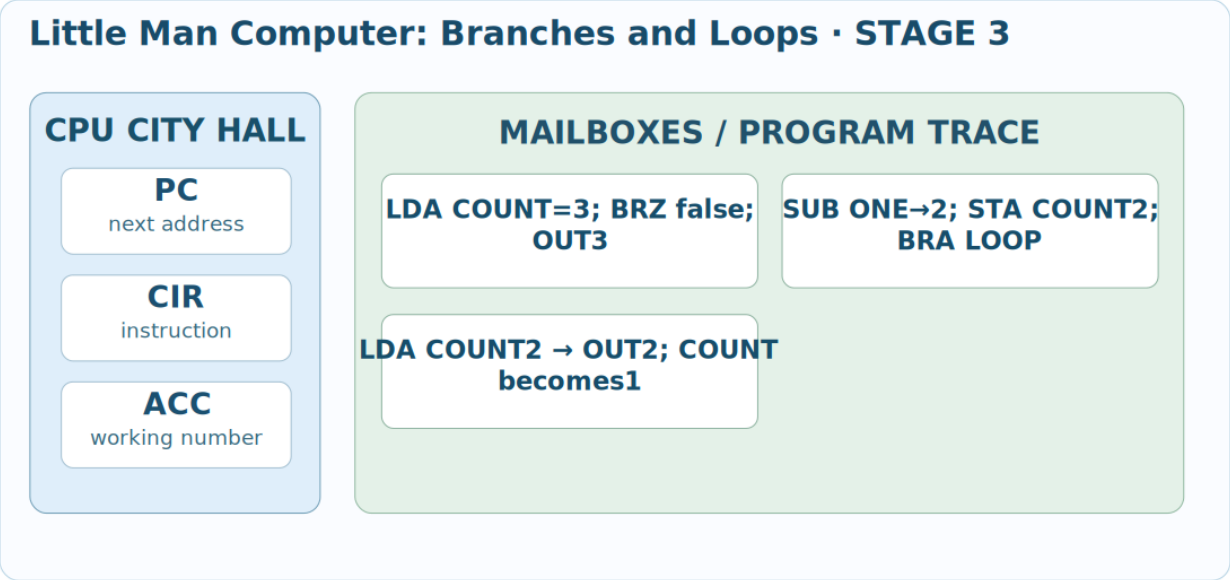



Little Man Computer: Branches and Loops · STAGE 2; LDA COUNT=3; BRZ false; OUT3; SUB ONE→2; STA COUNT2; BRA LOOP

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**

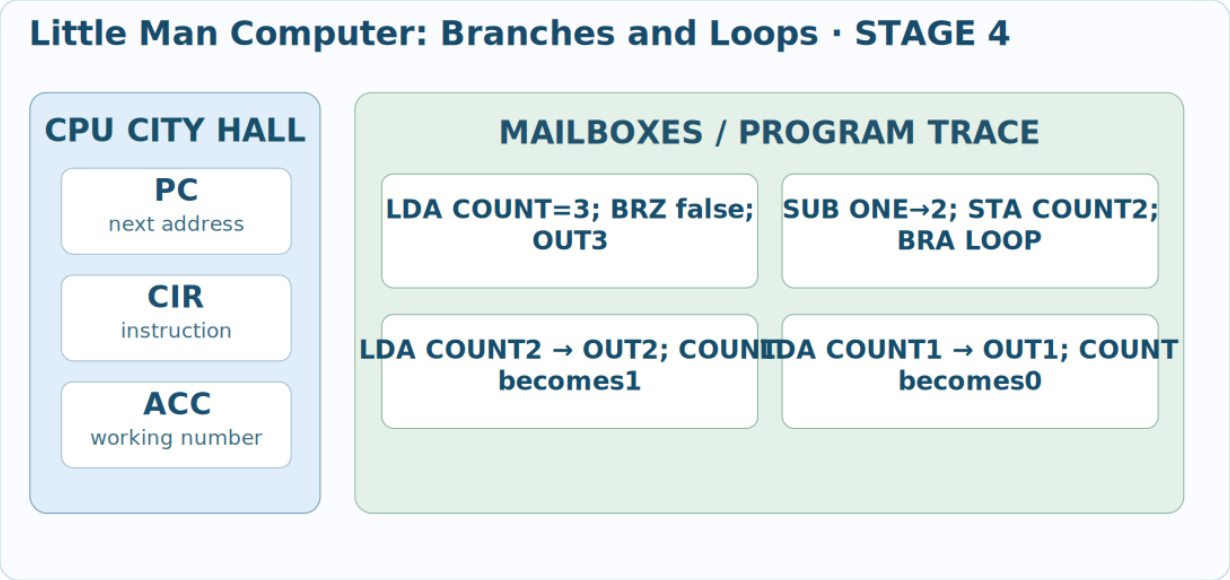



Little Man Computer: Branches and Loops · STAGE 3; LDA COUNT=3; BRZ false; OUT3; SUB ONE→2; STA COUNT2; BRA LOOP; LDA COUNT2 → OUT2; COUNT becomes1

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**

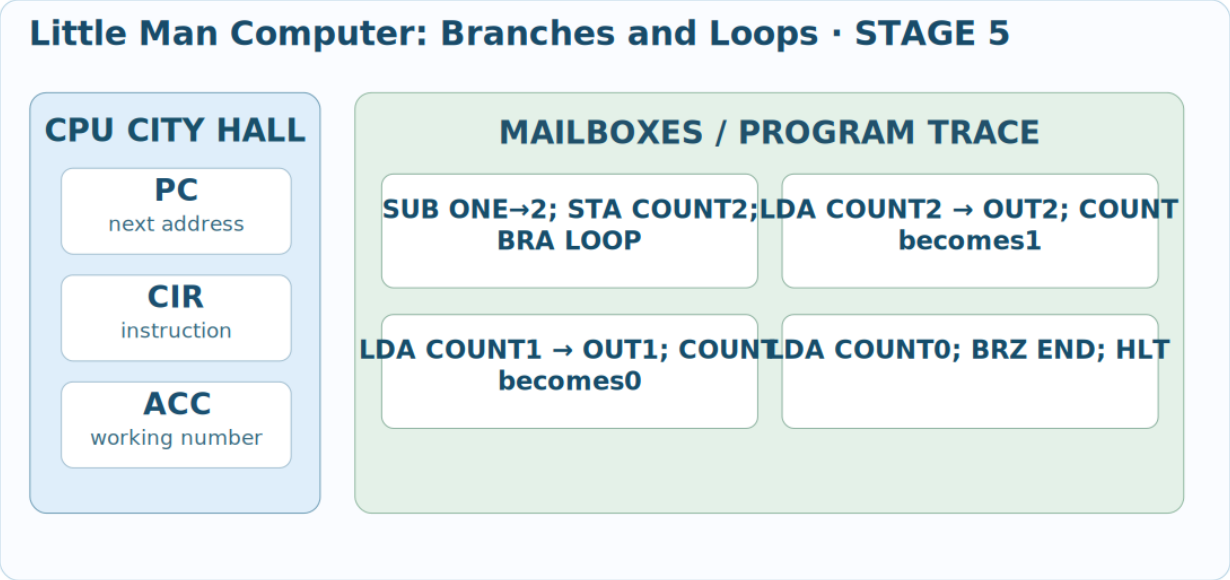



Little Man Computer: Branches and Loops · STAGE 4; LDA COUNT=3; BRZ false; OUT3; SUB ONE→2; STA COUNT2; BRA LOOP; LDA COUNT2 → OUT2; COUNT becomes1; LDA COUNT1 → OUT1; COUNT becomes0

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Little Man Computer: Branches and Loops · STAGE 5; LDA COUNT=3; BRZ false; OUT3; SUB ONE→2; STA COUNT2; BRA LOOP; LDA COUNT2 → OUT2; COUNT becomes1; LDA COUNT1 → OUT1; COUNT becomes0; LDA COUNT0; BRZ END; HLT

I DO — WORKED EXAMPLE

# Follow the worked method

### Start COUNT=2 instead of3. What gets printed and why?

### Follow the reasoning

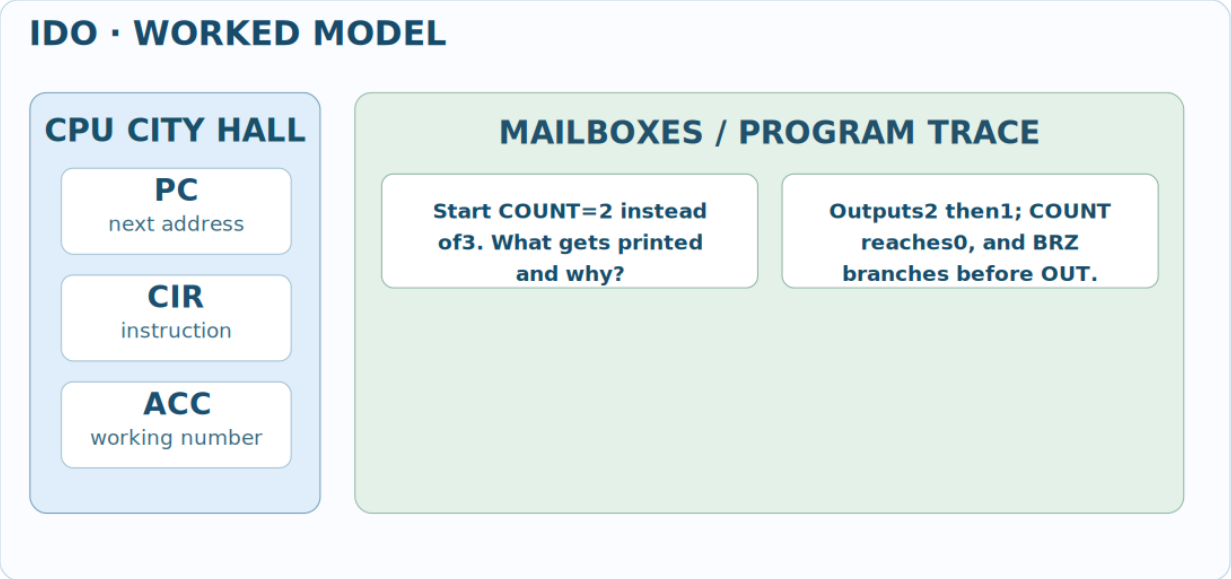
Outputs2 then1; COUNT reaches0, and BRZ branches before OUT.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Start COUNT=2 instead of3. What gets printed and why?

Outputs2 then1; COUNT reaches0, and BRZ branches before OUT.




IDO · WORKED MODEL; Start COUNT=2 instead of3. What gets printed and why?; Outputs2 then1; COUNT reaches0, and BRZ branches before OUT.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

In a separate program, COUNT starts4 and SUB TWO is used, TWO DAT2. Give outputs.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Trace COUNT DAT3 loop outputs and COUNT after each SUB. [6]

### Question 2

Change COUNT DAT3 to COUNT DAT2. State outputs and explain BRZ. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

A pupil replaces BRZ END with BRA END at LOOP. Explain effect and repair. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**BRZ END always jumps to END regardless of ACC.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

## GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

## Guided 1 — CHECK

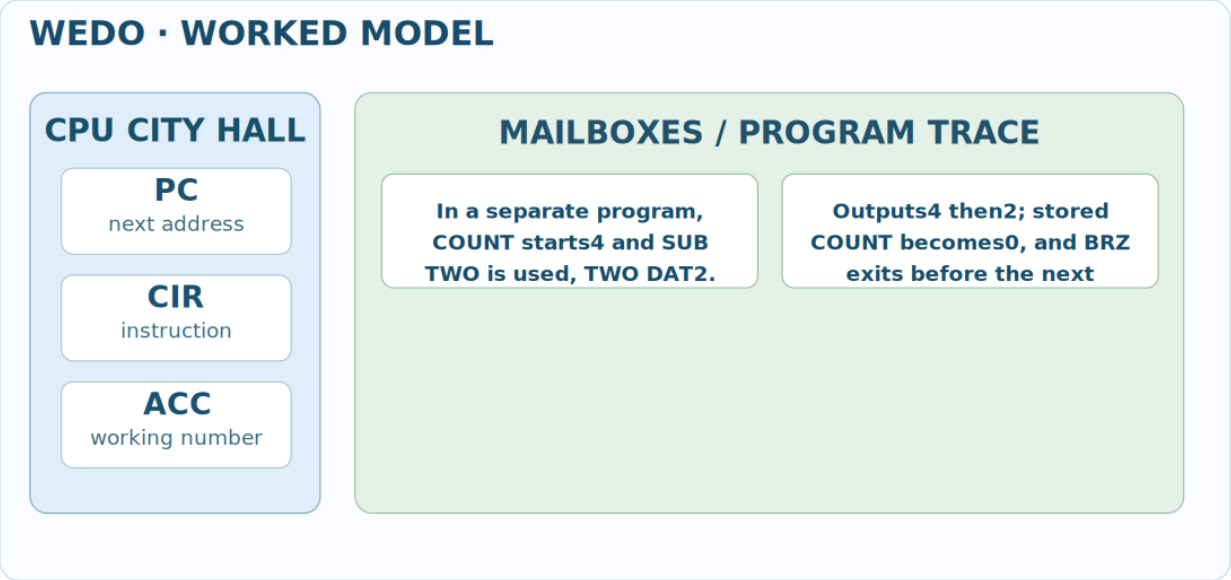
Follow the reasoning
Outputs4 then2; stored COUNT becomes0, and BRZ exits before the next OUT.

# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**In a separate program, COUNT starts4 and SUB TWO is used, TWO DAT2. Give outputs.**

Outputs4 then2; stored COUNT becomes0, and BRZ exits before the next OUT.




WEDO · WORKED MODEL; In a separate program, COUNT starts4 and SUB TWO is used, TWO DAT2. Give outputs.; Outputs4 then2; stored COUNT becomes0, and BRZ exits before the next OUT.